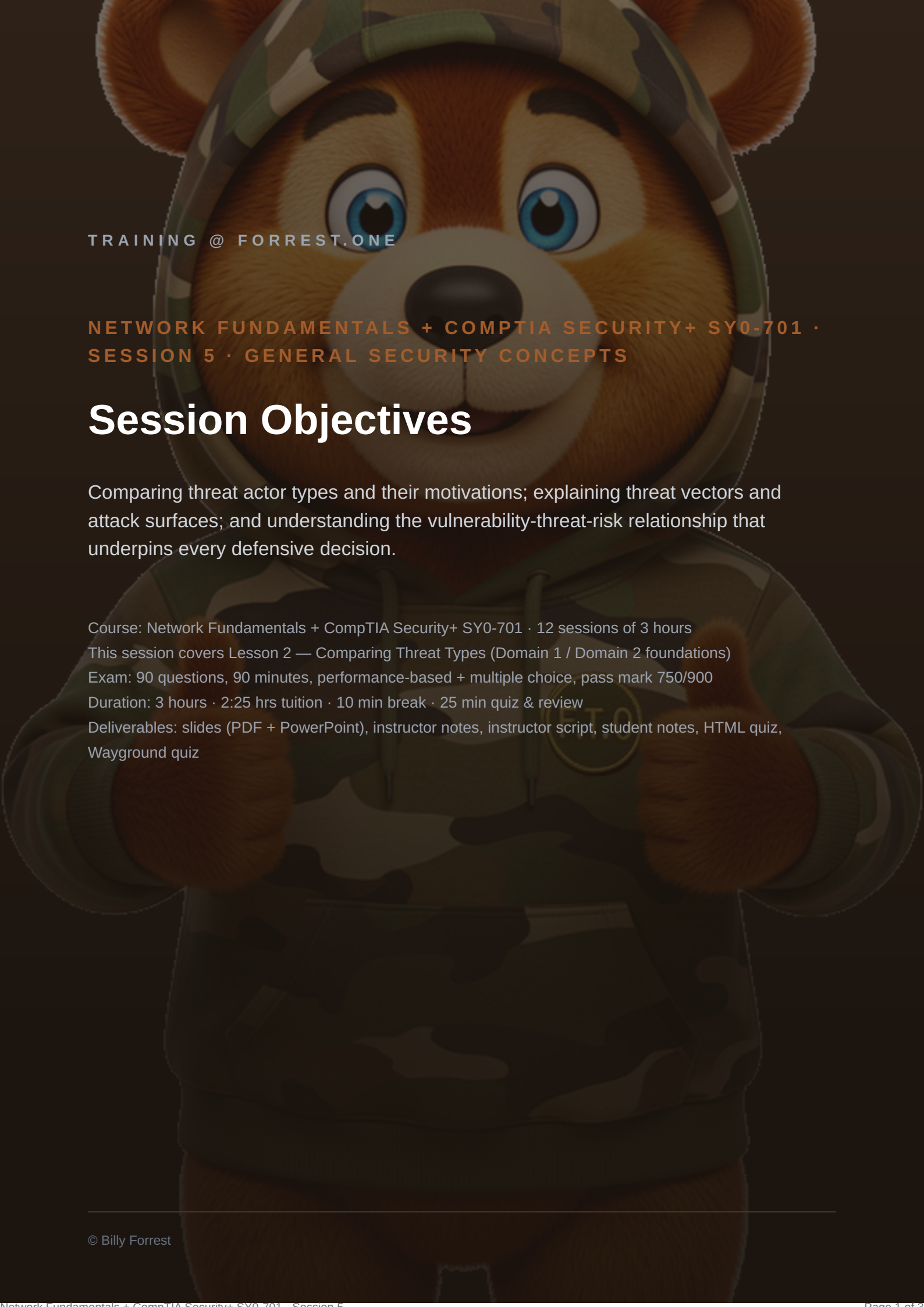




TRAINING @ FORREST.ONE

NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 5 · GENERAL SECURITY CONCEPTS

Session Objectives

Comparing threat actor types and their motivations; explaining threat vectors and attack surfaces; and understanding the vulnerability-threat-risk relationship that underpins every defensive decision.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

This session covers Lesson 2 — Comparing Threat Types (Domain 1 / Domain 2 foundations)

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Session Objectives — Threats, Attack Types & Threat Actors

This session puts faces to the threats behind every control in the course. It covers Lesson 2 of the Cyber Academy programme, Comparing Threat Types, and it now spans the whole of Domain 2 (Threats, Vulnerabilities and Mitigations, 22% of the exam): the actors, the attack surface, the vulnerabilities, the attacks themselves, and the mitigations that answer them.

Learning outcomes

By the end of the 3-hour session, candidates will be able to:

- Define vulnerability, threat and risk, and explain how they combine.
- Describe the attributes of threat actors: resources, sophistication, motivation and intent.
- Compare threat actor types: nation-state actors and APTs, organised crime, hacktivists, unskilled attackers, malicious and negligent insiders, shadow IT, and competitors.
- Explain threat actor motivations: financial, political or ideological, strategic, personal and ethical, plus the use of false flags to defeat attribution.
- Define the attack surface and the factors that expand it, including cloud, mobile, Internet of Things and remote work.
- Explain threat vectors, including human vectors such as the phishing family, social engineering, direct access, remote access, open service ports and the supply chain.
- Explain the main types of vulnerability: supply chain and software, application and web, operating system and hardware, and mobile, cloud and virtualisation.
- Describe malware by lifecycle and type, explain evasion techniques, and identify ransomware, spyware, rootkits and botnets.
- Explain password and credential attacks, from brute force to credential stuffing and offline hash cracking.
- Describe network attacks: injection, denial of service and amplification, Layer 2, DNS, rogue devices and on-path attacks, plus physical and cryptographic attacks.
- Apply the mitigations: segmentation, least privilege, need to know, separation of duties, hardening, patch management and isolation.
- Describe the attack chain: reconnaissance, vector, exploit, act.

CompTIA exam context

Item	Detail
Exam	Network Fundamentals + CompTIA Security+ SY0-701

Item	Detail
Objectives	2.1 Threat actors & motivations; 2.2 Threat vectors & attack surfaces; 2.3 Types of vulnerabilities; 2.4 Indicators of malicious activity; 2.5 Mitigation techniques
Domain 2 weight	Threats, Vulnerabilities and Mitigations, 22% (the domain behind this whole session)
Pass mark	750 on a scale of 100–900
Style	Scenario-based questions reward classification: name the actor, the vector, the weakness or the control

Chapter coverage

#	Chapter	What is covered
1	Know the Adversary	Vulnerability-threat-risk; the four attributes; motivations and intent; nation-state actors and APTs; organised crime; hacktivists, unskilled attackers and insiders; competitors and false flags
2	The Attack Surface	The surface and what expands it; common vectors; the phishing family; social engineering; network and file-based vectors; direct access, remote access and open services
3	Vulnerabilities	Supply chain and software; application and web; operating system and hardware; mobile, cloud and virtualisation
4	Malware and Credential Attacks	The malware lifecycle and its types; evasion; ransomware, spyware and rootkits; botnets and command and control; password attacks
5	Attacks on the Network	Injection; denial of service and amplification; Layer 2; DNS; rogue devices and on-path; physical and cryptographic attacks
6	Mitigation	Segmentation; access control; hardening and patch management; isolation strategies

How this session maps to the exam

Every activity and quiz question is tagged with the exam objective it tests (for example 2.2 Threat Vectors & Attack Surfaces). Using the session quiz to 70%+ before the exam is the fastest route to marks in this domain, which carries the second largest weighting in the paper.